6.2 False Refusal in Language Models

Testing for False Refusal The first test suite explicitly designed for evaluating false refusal (or "exaggerated safety") in LLMs was XSTest, introduced by Röttger et al. (2024). XSTest consists of 250 hand-written safe prompts across ten prompt types, as well as 200 contrasting unsafe prompts. Subsequent work has expanded on XSTest's scope, using LLMs to generate larger sets of safe test prompts. Specifically, Cui et al. (2024a) create OR-Bench as a collection of 80k "seemingly toxic" prompts across ten rejection categories. Similarly, An et al. (2024b) create PHTest, which consists of 3,260 "pseudo-harmful" prompts. Shi et al. (2024) create OKTest, comprising only 350 safe questions. Chehbouni et al. (2024) create a more specialised templated dataset for testing sociodemographic biases in false refusal.

Mitigating False Refusal False refusal mitigation methods can be generally categorised into training-free and training-based methods. Training-free methods are normally adaptive and query-specific, which leads to computational overhead during inference. Self-CD (Shi et al., 2024) applied contrastive decoding by inferencing twice on the same query with and without the system prompt. As a contemporary work, SCAN (Cao et al., 2024) constructed a classifier to adaptively decide whether the model should refuse a certain query, which is controlled by activation steering. The additional classifier requires additional memory usage and computation at the inference time, and the activation steering through subtraction is non-surgical. Training-based methods including Zheng et al. (2024) and Zhang et al. (2024) require training samples to calibrate the model and lack the flexibility for post-training adjusting for personalization. Our approach is training-free and prompt-agnostic, meaning the intervention can be carried out directly to the model weights and applied to any queries which leads to no additional inference cost.

6.3 Representation Editing

Representation editing involves steering the model behaviour by influencing the hidden representation of the model. Recent work has successfully demonstrated its success in controlling the truthfulness (Li et al., 2024) and sentiment (Rimsky et al., 2023; Turner et al., 2023). Zou et al. (2023a) proposed various techniques for finding representations of high-level concepts such as honesty and emotions. Arditi et al. (2024) focused on the refusal behaviour of the model and extracted a refusal direction from the activation stream of the transformer model. The resulting direction controls the general refusal behaviour regardless of whether it is a true or false one. Our work takes a more surgical and fine-grained approach to mitigate the false refusal without harming the other.

7 Conclusion

In this work, we proposed a surgical approach to mitigate the false refusal in language models via ablating a single vector from the activation stream of the transformer model. We ablated a false refusal vector extracted from the mean model activation difference when prompting the model with pseudo-harmful and harmless data. To avoid compromising model safety on harmful queries, we applied orthogonalization to remove the projection of the false refusal vector onto the true refusal vector. We also showed that fine-grained calibration of the safety guard can be achieved by partial orthogonalization and adjusting the strength of the projection removal. This enables more flexibility for adjusting the response compliance level to ambiguous and sensitive queries that are up to the user's judgment. The surgical characteristic of the ablation operation keeps the modified model's original general capabilities and requires no additional inference cost via direct model weight modification.

Limitations We show that the false refusal vector can be extracted using a small sample of pseudo-harmful and harmless queries. However, the quality of the vector can be further improved by constructing more diverse pseudo-harmful samples, as we mainly focused on using pseudo-harmful samples from OR-Bench-Hard (Cui et al., 2024b). A better data curation strategy and sampling technique could further improve the effectiveness of the vector, which we leave for future work. To select the refusal vector, we adopt the refusal score metrics from previous work (Arditi et al., 2024) which looks at the refusal-related tokens at the first token position. Such evaluation metrics can be further improved, such as considering sequence probabilities, for better refusal vector selection.

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